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PERFORMANCE SPECIFICATION

VALVE, BYPASS, FUEL, MECHANICAL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mechanical fuel bypass valve, herein referred to as "the assembly".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4820

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52557 - Fuel Oil, Diesel; For Posts, Camps and Stations.

DEPARTMENT OF DEFENSE

MIL-C-7024 - Calibrating Fluids, Aircraft Fuel System Components.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12324736-1 - Electromechanical Fuel System Assembly.
12324795 - Valve, Bypass, Fuel, Mechanical.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for the valve assemblies. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.4.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Corrosion control. The assembly shall be fabricated from materials that are inherently corrosion resistant or are permanently treated to prevent forms of corrosion and deterioration which affect system performance (see 4.4.1).

3.3 Design and construction. The assembly shall conform to the manufacturer's design and construction except for the envelope dimensions; port hole dimensions, locations, and depth; lever swing radius and location; and mounting dimensions which shall conform to Drawing 12324795 and other applicable drawings.

3.3.1 Interchangeability. All assemblies shall be interchangeable as to fit, form, and function with other manufacturer's assemblies having the same military part number.

3.3.2 Weight. The weight of the assembly shall be from 0.73 to 1.04 kilograms (kg) (1.6 to 2.3 pounds) (see 4.4.3).

3.4 Performance. Unless otherwise specified (see 6.2), performance test shall be conducted with the valve plumbed as shown in figure 1.

3.4.1 Valve schedule. The assembly shall meter servo fuel flow (PR) and nozzle fuel flow (PN) based on A-A-52557, Grade DF-2 diesel fuel oil at a specific gravity of 0.84 and a temperature of 15 to 21 degrees Celsius (°C) [60 to 70 degrees Fahrenheit (°F)]. PR and PN

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schedule shall be flow tested using MIL-C-7024, type II calibrating fluid. The DF-2 fuel flow schedule on Drawing 12324795 may be converted to a schedule for MIL-C-7024, type II calibrating fluid by using the following formula based on specific gravity (Sp. Gr.) (see 4.4.4.1):

$$\text{Fuel flow rate MIL-C-7024, type II} = \left(\sqrt{\frac{0.77}{0.84}}\right) \times \text{Fuel flow rate DF-2 (pph).}$$

[pounds per hour (pph)] [kilogram per hour (kg/h)]

3.4.1.1 PR cracking angle and flow schedule. The PR port shall begin to flow (crack open) at an operating lever angle of 7 to 13 degrees. The PR cracking angle test shall be performed with P1 inlet minus PR port outlet pressure equal to 19 psid.

a. Metered flow. The assembly shall meter PR flow with 19 psid (P1 inlet minus PR outlet) and P1 equal to 500 psig, in accordance with the following:

(1) Minimum 67 pph when the PN valve system cracks open.
(2) As specified in the metered flow table, section 3.4.1.2.

3.4.1.2 PN cracking angle and flow schedule. The PN port shall begin to flow (crack open) at an operating lever angle of 11 to 19 degrees. The PN cracking angle test shall be performed with P1 minus PN port pressure equal to 150 psid.

a. Metered flow. The assembly shall meter PR flow with 150 psid (P1 inlet minus PN outlet) and P1 equal to 500 psig, as specified in the metered flow table I.

TABLE I. Metered flow (using DF-2 diesel fuel).

	PR flow			PN flow		
Position	Minimum	Actual	Maximum	Minimum	Actual	Maximum
30°	79		100	132		228
40°	79		100	252		348
50°	79		100	372		468
60°	79		100	460		500
70°	79		100	460		500

3.4.2 Proof pressure. Assembly components shall show no evidence of failure, permanent set, or damage when subjected to a gage pressure of 10.34 megaPascals (MPa) [1500 pounds per square inch (psi)] (see 4.4.4.2).

3.4.3 Leakage. Maximum PR or PN port leakage and PR to PN cross port leakage at closed position (handle = 0°) shall be 0.5 milliliters per minute (mL/min). There shall be no evidence of external leakage (see 4.4.4.3).

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3.4.4 Handle torque. The torque required to operate the assembly through its rotational range from either direction (0° to 70°, 70 to 0°) shall not exceed 0.9 Newton-meter (N-m) (8 inch-pounds) (see 4.4.4.4).

3.5 Environmental conditions.

3.5.1 Temperature. The assembly shall meet the requirements of 3.4.3 (port leakage only) during exposure, and 3.4 after exposure to temperatures ranging from -51 to +121°C (-60 to +250°F) (see 4.5.1).

3.5.2 Fungus. The assembly shall not support fungus growth and shall meet the performance requirements of 3.4 (see 4.5.2).

3.5.3 Vibration. The assembly, when fixtured as an integral part of Hydromechanical Fuel Control Assembly 12324736-1, shall meet the requirements of 3.4.3 (port leakage only) during exposure, and 3.4 after exposure to the peak vibration envelope defined in table II and figure 2 in each of three mutually perpendicular axes, at a logarithmic sweep rate of 15 minutes from 5 to 500 to 5 Hertz (Hz) (see 4.5.3).

TABLE II. Test spectra for vibration test.

Segment no.	Frequency (Hz)	Function (D A)	Level (mil pk-pk)	(g-pk)
1	5	A	-	1.00
2	25	D	30.00	-
3	51	A	-	4.00
4	300	A	-	1.70
5	500			1.70

Function Key: D = displacement
A = acceleration

3.5.4 Shock.

3.5.4.1 Basic shock. The assembly shall meet the requirements of 3.4 after exposure to shock impulses of 20 ± 2 g for 18 ± 1.8 milliseconds (ms) half sine wave in each direction of three mutually perpendicular axes (see 4.5.4.1).

3.5.4.2 Gun firing shock. The assembly shall meet the requirements of 3.4 after exposure to shock impulses in each direction of three mutually perpendicular axes shown in figure 5. Gun firing shock levels shall be defined by the shock response spectra shown in figures 6,7, and 8 for the vertical, axial, and lateral directions respectively (see 4.5.4.2).

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3.5.4.3 Ballistic shock. The assembly shall meet the requirements of 3.4 after exposure to shock impulses in each direction of the three mutually perpendicular axes shown in figure 5. Ballistic shock levels shall be defined by the shock response spectra shown in figures 9, 10, and 11 for the vertical, axial, and lateral directions respectively (see 4.5.4.3).

3.5.5 Sand and dust. The assembly shall meet all of the performance requirements of 3.4 after exposure to a sand and dust environment as specified in 4.5.5.

3.5.6 Operation. The assembly shall be capable of extended operation with the handle in the full closed position (0°). The torque required to rotate the handle, after extended operation with the handle in the full closed position (0°), shall not be more than 1.35 N-m (12 inch-pounds) (see 4.5.6).

3.6 Identification marking. Identification marking shall include, as a minimum, the following information permanently marked on the exterior of the assembly (see 4.6):

a. Manufacturer's identification.
b. Serial number
c. Military part number
d. Federal stock number
e. Manufacturer's part number
f. Date of manufacture
g. Contract number
h. U.S.

3.7 Parts. Contractor shall supply a list and necessary data for replacement and spare parts. The contractor shall also assure that parts can be purchased using the contractors product drawings(s) and that the hardware can be built from the drawing package.

3.8 Workmanship. The assembly shall be free of defects which may adversely affect the serviceability or appearance of the valve. Particular attention shall be given to the neatness and thoroughness of welding, brazing, marking, plating, painting, and freedom of parts from burrs and sharp edges (see 4.7).

4. VERIFICATION

Caution: Do not exceed the required test pressure differential across ports.

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.2).
   1. Preproduction inspection (see 4.2.1).
   2. Initial production inspection (see 4.2.2).
b. Conformance inspections (see 4.3).
   1. Examination (see 4.3.1).
   2. Tests (see 4.3.2).

4.1.1 Accuracy of inspection equipment. Unless otherwise specified (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to ensure that supplies and services conform to contract requirements. Inspection equipment shall be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance, except as follows:

a. Handle position to ±1° of rotation.
b. Fuel flow rate to 0.5 % of actual.
c. P1 pressure to ±5 pounds per square inch gage (psig) ±35 kiloPascal (kPa).
d. P1 - PR delta pressure to ±0.5 psi ±3.5 kPa.
e. P1 - PN delta pressure to ±2 psi ±14 kPa.

4.2 Preproduction.

4.2.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of one or more assemblies. Preproduction inspection shall consist of inspection as specified in table II.

4.2.1.1 Preproduction requirements. When specified (see 6.2), the preproduction testing of table III shall be performed. Performance testing of 4.4.4.1, 4.4.4.2, 4.4.4.3.1, and 4.4.4.3.2, shall be performed before and after environmental tests of 4.5.

Handle Position (°)	PR (pph-DF-2) (kgph)	PN (pph-DF-2)(kgph)
30	78-102 (35-46)	122-238 (55-108)
40	78-102 (35-46)	242-358 (110-162)
50	78-102 (35-46)	362-478 (164-217)
60	78-102 (35-46)	450-510 (204-231)
70	78-102 (35-46)	450-510 (204-231)

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TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article		CI (100%)		Control
			Pre-prod.	Initial prod.	Examination	Tests	
Materials, design, and construction	3.2 thru 3.3	4.4.1	X	X			
Defects	3.3., 3.3.1, 3.6, and 3.7	4.4.2	X	X	X		X
Weight	3.3.1	4.4.3	X	X			X
Performance	3.4	4.4.4					
Valve schedule	3.4.1	4.4.4.1					
PR schedule	3.4.1.1	4.4.4.1	X	X		X	X
PN schedule	3.4.1.2	4.4.4.1	X	X		X	X
Proof pressure	3.4.2	4.4.4.2	X	X		X	X
Leakage	3.4.3	4.4.4.3	X	X		X	X
Port leakage		4.4.4.3.1	X	X		X	X
Cross port leakage		4.4.4.3.2	X	X		X	X
Handle torque	3.4.4	4.4.4.4	X	X		X	X
Environmental	3.5	4.5					
Temperature	3.5.1	4.5.1					
High temperature	3.5.1	4.5.1.1	X	X			X
Low temperature	3.5.1	4.5.1.2	X	X			X
Fungus	3.5.2	4.5.2	X	X			X
Vibration	3.5.3	4.5.3	X	X			
Control		4.5.3.1					X
Shock	3.5.4	4.5.4					
Basic shock	3.5.4.1	4.5.4.1	X	X			
Gun firing shock	3.5.4.2	4.5.4.2	X	X			
Ballistic shock	3.5.4.3	4.5.4.3	X	X			
Sand and dust	3.5.5	4.5.5	X	X			
Operation cycle	3.5.6	4.5.6	X	X			X
ID and marking	3.6	4.6	X	X		X	X
Workmanship	3.7	4.7	X	X		X	X
Packaging		5		X			X

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability out of tolerance (see 3.3).	SIE 1/
103	Workmanship, faulty affecting appearance(see 3.8).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Marking, improper (see 3.6).	Visual
203	Workmanship, faulty, affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), two selector valve assemblies from the first 100 selector valves produced shall be selected for initial production inspection. This inspection shall include the examinations of 4.3.2 (see table IV) and the tests of 4.4 (see table III).

4.2.2.1 Initial production requirements. When specified (see 6.2), the initial production testing of table III shall be performed. Performance testing of 4.4.4.1, 4.4.4.2, 4.4.4.3.1, and 4.4.4.3.2, shall be performed before and after environmental tests of 4.5.

4.2.2.2 Control test requirements. Assemblies undergoing control testing shall be subjected to the performance tests of 4.4.1, 4.4.2, 4.4.3 and 4.4.4 before and after each environmental test. The test points and flow limits for assembly schedule flow (see 4.4.4.1) acceptance shall be as follows:

Handle Position (°)	PR (pph-DF-2)	PN (pph-DF-2)
30	78-102	122-238
40	78-102	242-358
50	78-102	362-478
60	78-102	450-510
70	78-102	450-510

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.3.1 and the tests of 4.3.2.

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4.3.1 Examination. Each selector valve assembly shall be examined for the defects specified in table IV.

4.3.2 Acceptance tests (100 percent). Each selector valve assembly shall be subjected to the inspections specified in table III. Failure of any inspection shall be cause for rejection.

4.3.3 Control tests. Unless otherwise specified (see 6.2), control tests shall be conducted on one assembly from each lot 500 units consecutively produced, except that not more than one test shall be perfomed in a 6-month period, nor less than one test in a 12 month period. Assemblies undergoing control sample testing shall be subjected to the tests specified in table III. Performance testing of 4.4.4.1, 4.4.4.2, 4.4.4.3.1 and 4.4.4.3.2 shall be performed before and after completion of environmental tests of 4.5.

4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.3.1 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection.

4.4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: 23 ±10°C (73 ±18°F)
b. Pressure: 597 to 775 mm (23.5 to 30.5 inches) of mercury
c. Relative humidity: 50 ±30%
d. Fuel temperature: 27 ±5.6°C (80 ±10°F)

4.4.3 Weight. The assembly shall be weighed to verify conformance to 3.3.2.

4.4.4 Performance. Unless otherwise specified (see 6.2), performance tests shall be conducted with MIL-C-7024, type II calibrating fluid. Flow measurements shall be adjusted to account for the differences in specific gravity between the test fluid and design fluid A-A-52557, grade DF-2.

4.4.4.1 PR and PN cracking angle and flow schedule. A test to determine what lever angle the PR and PN ports crack open shall be accomplished. The crack open tests may be performed either with the PR and PN ports open or installed in a circuit that has flow meters in the outlet lines with up to 500 psig at P1 (inlet). For either test method, the required differential pressure of 19 psid (P1 inlet minus PR outlet) and 150 psid (P1 minus PN) shall be maintained. Crack open is described as the point where a steady flow stream begins or 10 pph on the flow meter is observed. The assembly shall be tested to demonstrate metered flow in accordance with

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3.4.1.1 and 3.4.1.2. The metered flow test shall be performed with calibrated flow meters in the PR and PN outlet lines. P1 inlet pressure for the flow test shall be adjusted to 500 psig while maintaining the required pressure differential of 19 psid across P1 to PR ports and 150 psid for the P1 to PN ports.

4.4.4.2 Proof pressure. To determine conformance to 3.4.2, the assembly shall be set to the closed position (0°), and the long tube "B" nut fitting capped. The short tube "B" nut fitting (P1) and PN and PR ports shall be simultaneously pressurized with MIL-C-7024, type II, calibrating fluid to a gage pressure of 10.34 MPa (1500 psi) and shall be held for 2 minutes. There shall be no evidence of external leakage.

4.4.4.3 Leakage.

4.4.4.3.1 Port leakage. To determine conformance to 3.4.3, the assembly shall be tested for port leakage with the handle in the full closed (0°) position, and the long tube "B" nut fitting capped. The short tube "B" nut fitting shall be pressurized with MIL-C-7024, type II, calibrating fluid to a gage pressure of 2.76 MPa (400 psi) and 0.69 MPa (100 psi) for one minute at each pressure. The PR and PN leakage rates shall be measured during the one minute at each pressure. Leakage shall not exceed 0.5 mL/min from either port.

4.4.4.3.2 Cross port leakage. To determine conformance to 3.4.3, the assembly shall be tested for cross port leakage with the handle in the full closed (0°) position, and the long tube "B" nut fitting capped. The short tube "B" nut fitting and PR port shall be pressurized to 0.21 MPa (30 psi) with MIL-C-7024, type II calibrating fluid and shall be held for one minute. Measure PN leakage rate during the one minute. Leakage shall not exceed 0.5 mL/minute.

4.4.4.4 Handle torque. To determine conformance to 3.4.4, handle torque shall be measured after the assembly has been exposed to MIL-C-7024, type II calibrating fluid so that the internal components are wet. The assembly shall not be pressurized while measuring handle torque. Handle torque shall be measured in both directions, from closed to open (0° to 70°) and from open to closed (70° to 0°).

4.5 Environmental conditions. Preproduction and initial production assemblies shall be subjected to the performance tests of 4.4.4.1 through 4.4.4.4 after each environmental test.

4.5.1 Temperature.

4.5.1.1 High temperature. To determine conformance to 3.5.1, the assembly shall be rigged at full closed (0°), supplied with MIL-C-7024, type II calibrating fluid at a gage pressure of 1.03 MPa (150 psi), and subjected to a high temperature test in accordance with the apparatus

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and preparation requirements of MIL-STD-810, Method 501.3 as follows. The assembly shall be exposed to an ambient temperature of 121°C (250°F) for one hour. During the test, the PN and PR ports shall be monitored for leakage.

4.5.1.2 Low temperature. To determine conformance to 3.5.1, the assembly shall be rigged at full closed (0°), supplied with MIL-C-7024, type II calibrating fluid at a gage pressure of 1.03 MPa (150 psi), and subjected to a low temperature test in accordance with apparatus and preparation requirements of MIL-STD-810, Method 502.3 as follows. The assembly shall be exposed to an ambient temperature of -51°C (-60°F) for one hour. During the test, the PN and PR ports shall be monitored for leakage.

4.5.2 Fungus. To determine conformance to 3.5.2, the assembly shall exhibit a fungus growth no greater than Grade 0 of MIL-STD-810, Method 508.4, Section I.

4.5.3 Vibration. To determine conformance to 3.5.3, the assembly shall be fixtured as an integral part of Hydromechanical Fuel Control Assembly 12324736-1 and plumbed with the two plugged fittings shown in figures 12A and 12B. The fixtured valve assembly shall be tested in accordance with the apparatus and preparation requirements of MIL-STD-810, Method 514.4, to the levels specified in table I and figure 2. These levels shall be applied to the base of the Hydromechanical Fuel Control Assembly as noted in figure 3. The valve assembly shall be pressurized with MIL-C-7024, type II, calibration fluid with the handle in the full closed (0°) position during exposure. The valve assembly tube (P1) shall be pressurized to a gage pressure of 1034.25 ±68.95 kilopascals (kPa) (150 ±10 psi). The PN and PR ports shall be vented to atmosphere. A 7.5 minute logarithmic sweep shall be conducted in each of three mutually perpendicular axes (see figures 3 and 4), from 5 to 500 Hz, to identify up to four of the most severe resonant frequencies. A resonant dwell shall then be conducted for 20 minutes at each of the resonant frequencies. The balance of the 120 minutes vibration exposure time in each axis shall be spent cycling the assembly from 5 to 500 to 5 Hz. The fixtured assembly may contain modified hoses and fittings for the purpose of pressurizing and measuring valve port leakage during exposure. The loose hoses, normally connected to the other engine components, shall be secured or removed prior to exposure. During exposure to vibration, the PR and PN ports of the valve assembly shall be monitored for leakage.

4.5.3.1 Vibration control test. The assembly shall be subjected to the tests of 4.5.3, except that the dwell time shall be 12.5 minutes per resonance and the total exposure time shall be 75 minutes per axis.

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4.5.4 Shock.

4.5.4.1 Basic shock. To determine conformance to 3.5.4.1, the assembly shall be mounted and testing shall be conducted using the installed interface configuration for its intended application, or equivalent and plumbed using the two plugged fittings shown in figures 12A and 12B. The assembly shall be tested in accordance with the apparatus and preparation requirements of MIL-STD-810, Method 516.4, as follows. The assembly shall be pressurized to a gage pressure of 1.03 MPa (150 psi) with MIL-C-7024, type II, calibrating fluid, and subjected to three half sine wave shock pulses applied in both directions along each of three mutually perpendicular axes (see figure 5). Pulse amplitude shall be 20 $\pm$2 g with a duration of 18 $\pm$1.8 ms. During exposure, the assembly shall be monitored for external leakage.

4.5.4.2 Gun firing shock. To determine conformance to 3.5.4.2, the assembly shall be mounted and testing shall be conducted using the installed interface configuration for its intended application, or equivalent and plumbed using the two plugged fittings shown in figures 12A and 12B. The assembly shall be tested in accordance with the apparatus and preparation requirements of MIL-STD-810, Method 516.4 as follows. The assembly shall be pressurized to a gage pressure of 1.03 MPa (150 psi) with MIL-C-7024, type II, calibrating fluid, and subjected to shock impulses meeting the Gun Firing Shock Response Spectra defined in figures 6, 7, and 8 for the vertical, axial, and lateral directions, respectively. Three shocks in both directions of each of the three axes shall be applied. The shock response shall be measured in the axis of shock at the short tube "B" nut, at the plans shown in figure 5, and the response shall be of equal or greater amplitude than that shown for each axis (figures 6, 7, and 8). During exposure, the assembly shall be monitored for external leakage.

4.5.4.3 Ballistic shock. To determine conformance to 3.5.4.3, the assembly shall be mounted and testing shall be conducted using the installed interface configuration for its intended application, or equivalent and plumbed using the two plugged fittings shown in figures 12A and 12B. The assembly shall be tested in accordance with the apparatus and preparation requirements of MIL-STD-810, Method 516.4 as follows. The assembly shall be pressurized to gage pressure of 1.03 MPa (150 psi) with MIL-C-7024, type II, calibrating fluid, and subjected to shock impulses meeting the Gun Firing Shock Response Spectra defined in figures 9, 10, and 11 for the vertical, axial, and lateral directions, respectively. Three shocks in both directions of each of the three axes shall be applied. The shock response shall be measured in the axis of shock at the short tube "B" nut, at the plans shown in figure 5, and the response shall be of equal or greater amplitude than that shown for each axis (figures 9, 10, and 11). During exposure, the assembly shall be monitored for external leakage.

4.5.5 Sand and dust. To determine conformance to 3.5.5, the assembly shall be subjected to a sand and dust environment for 10 hours. All ports shall be capped and the assembly rigged in the full closed position (0°). A stream of dust-laden air of 0.3 $\pm$0.2 gram per cubic foot density

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shall be directed at the valve shaft at a velocity of 900 ±300 feet per minute. A test chamber temperature of 27 ±5°C (80 ±10°F) and a relative humidity of 22 percent maximum shall be maintained during the test. Lever torque shall be recorded before and after the test.

4.5.6 Operation cycle. To determine conformance to 3.5.6, the assembly shall be supplied A-A-52557, grade DF-2 diesel fuel oil filtered through a 10 micron filter and delivered by a gear type pump. Refer to figure 13 for the plumbing circuit used in this test. With the assembly in the fully closed (0°) position, P1 gage pressure shall be set to 5.17 MPa (750 psi) and through fuel flow shall be 800 ±300 pph. Pressure drop across both ports at a handle position of 0° shall be 1.03 mPa (150 psi). The following cycle shall be repeated for a total of 15 times (360 hours):

a. Activate pump with handle closed (0°).
b. Deactivate pump after 4 hours.
c. Measure handle torque (0° to 70°, 70° to 0°).
d. Allow pump to remain idle for 20 hours with handle closed (0°).

4.6 Identification and marking. The pump assemblies shall be visually examined to verify conformance to the identification and marking requirements of 3.6.

4.7 Workmanship. Workmanship examinations shall be performed, following phases of the assembly process to verify conformance with 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly covered by this specification is intended for use on the AGT1500 engine.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. If first article inspection is required (see 3.1).
d. If inspection equipment should be maintained other than as specified (see 4.1.1).
e. If preproduction inspection is required (see 4.2.1).
f. If initial production inspection is required (see 4.2.2).
g. If control tests are to be conducted (see 4.3.3)
h. If inspection conditions should be other than as specified (see 4.4.2).
i. If performance tests should be conducted with alternate fuels (see 4.4.4).
j. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

AGT1500 engine
Battle override valve
Diesel
Fluid Control
Nozzle flow
Servo flow

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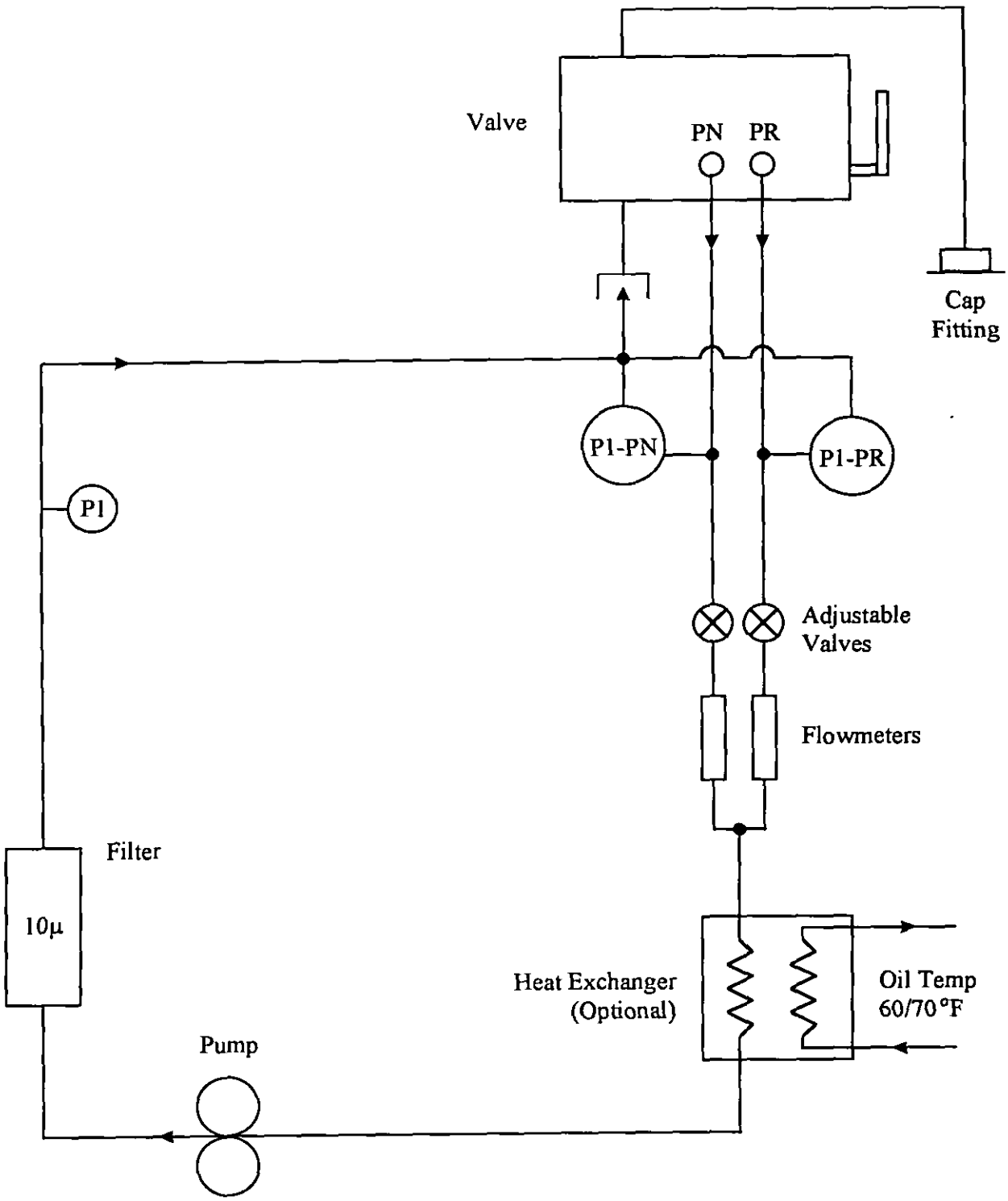


FIGURE 1. Flow test schematic.

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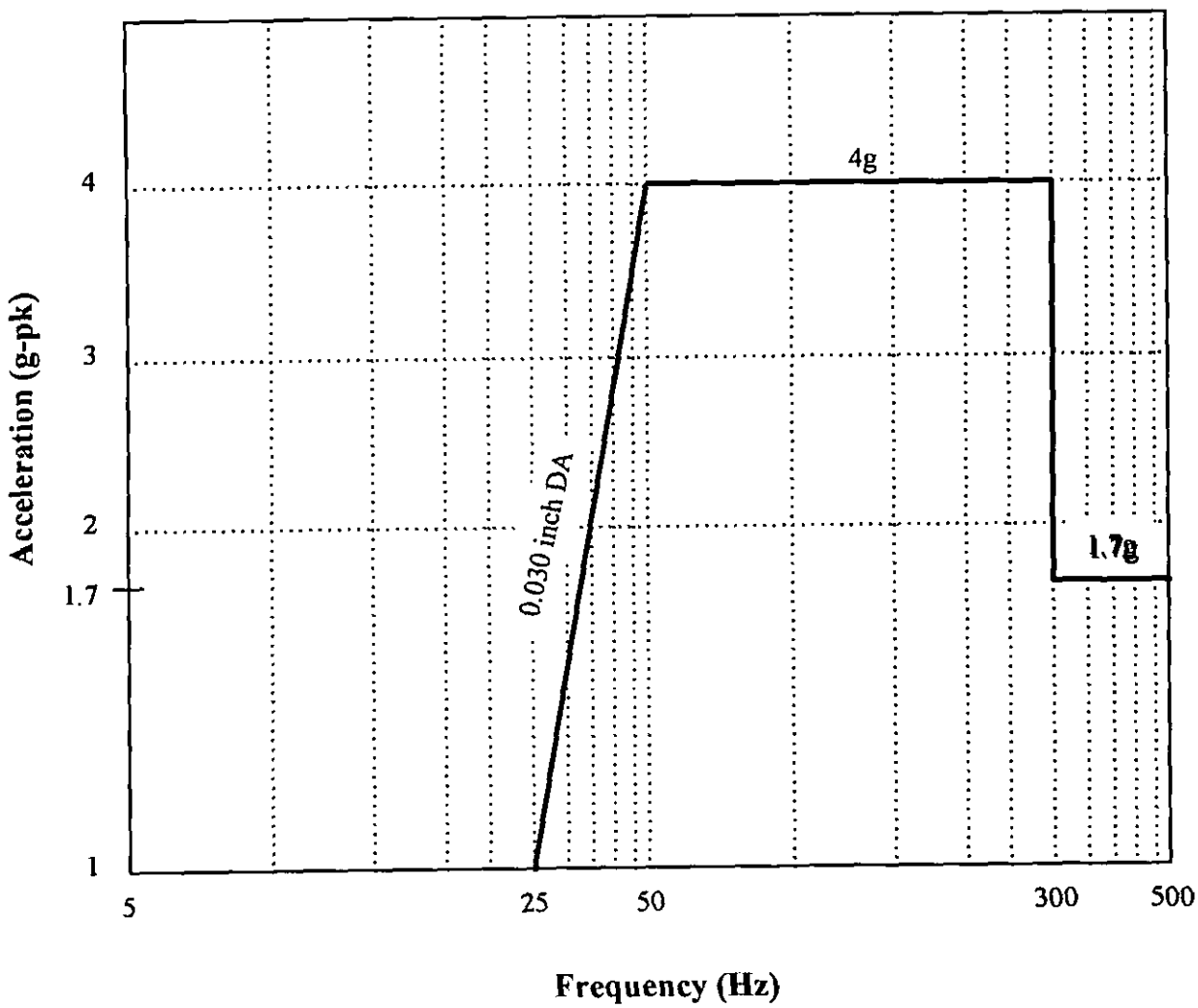


FIGURE 2. Vibration test spectrum.

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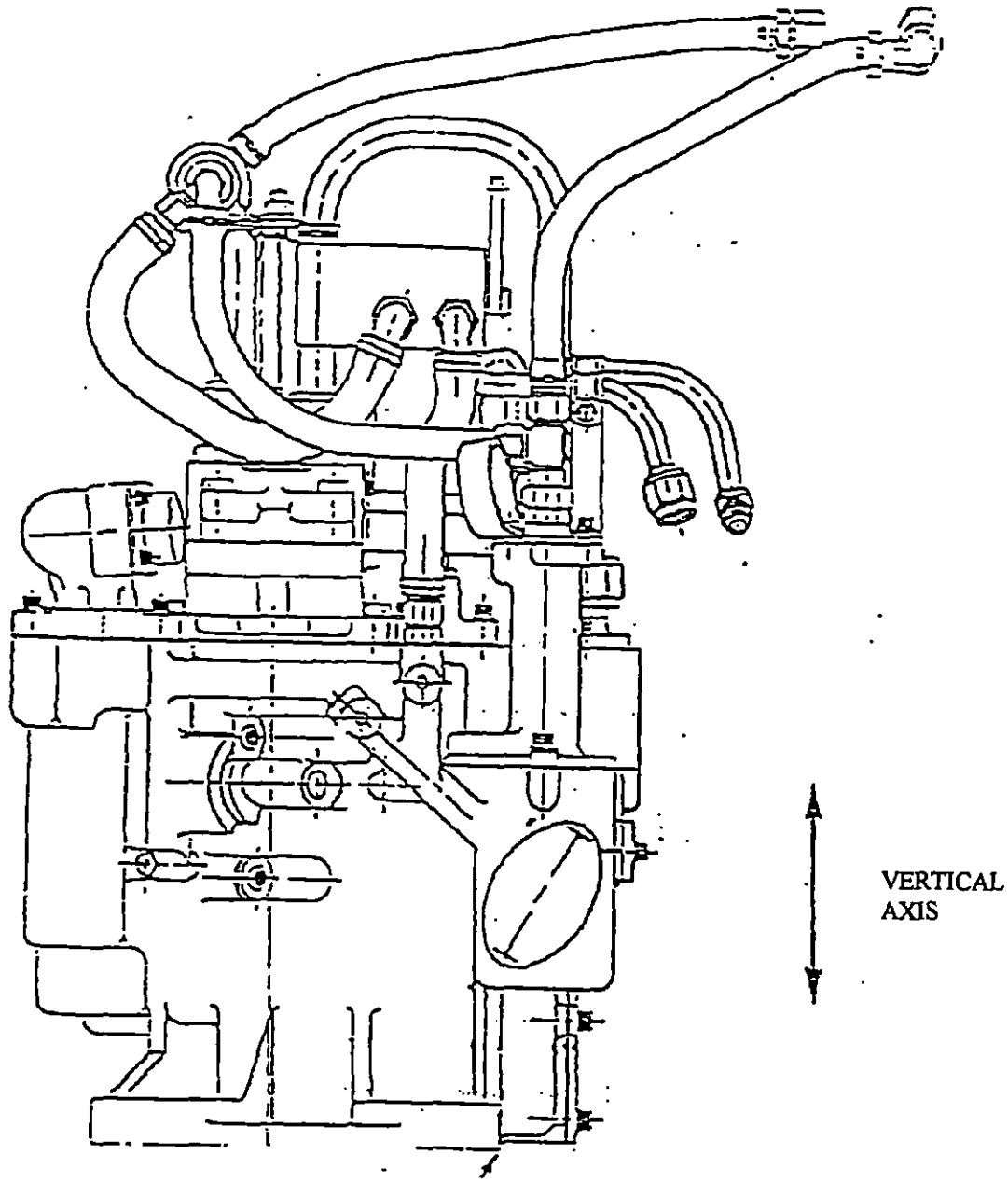


FIGURE 3. Vertical vibration test axis.

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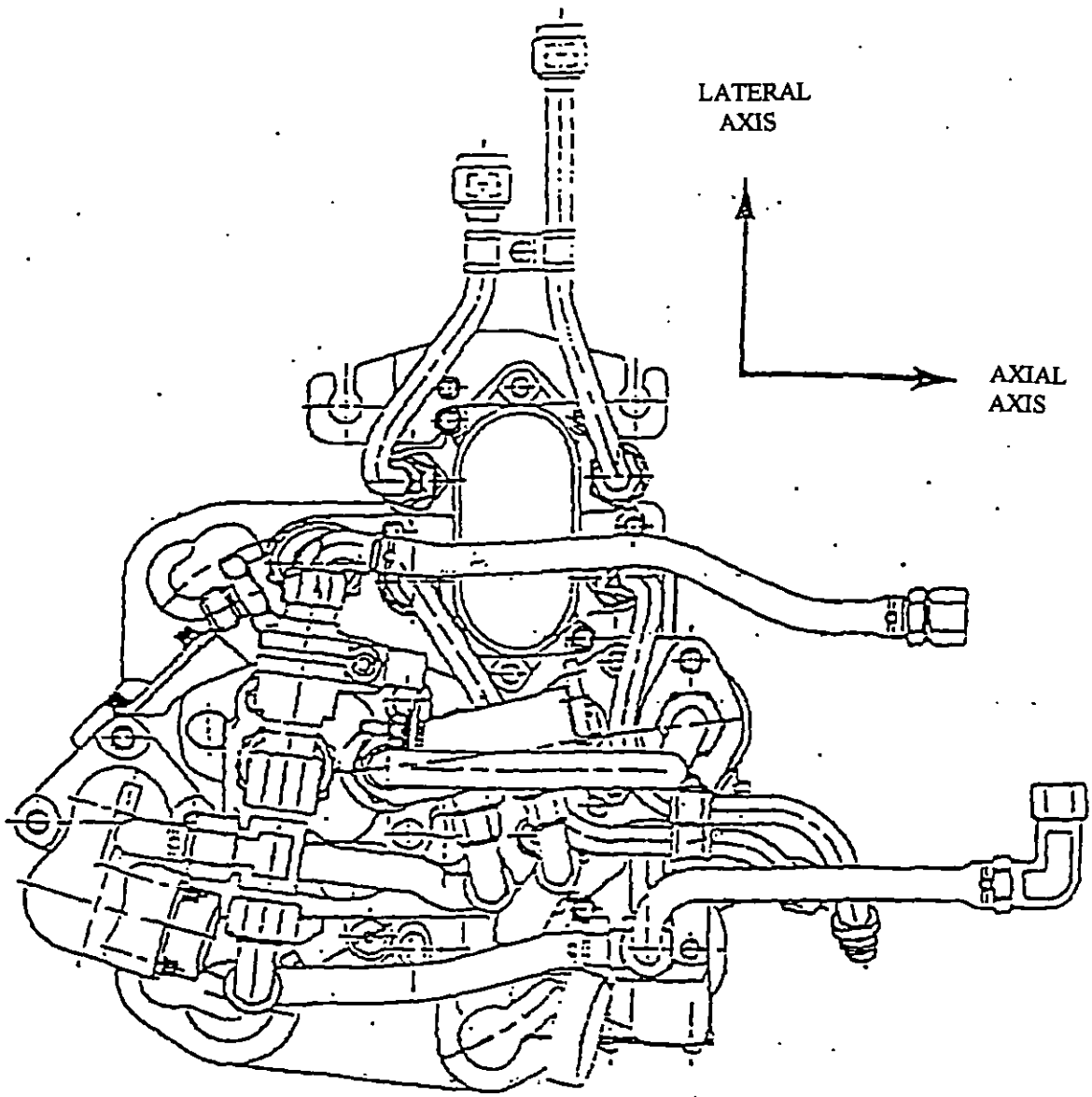


FIGURE 4. Lateral and axial vibration test axes.

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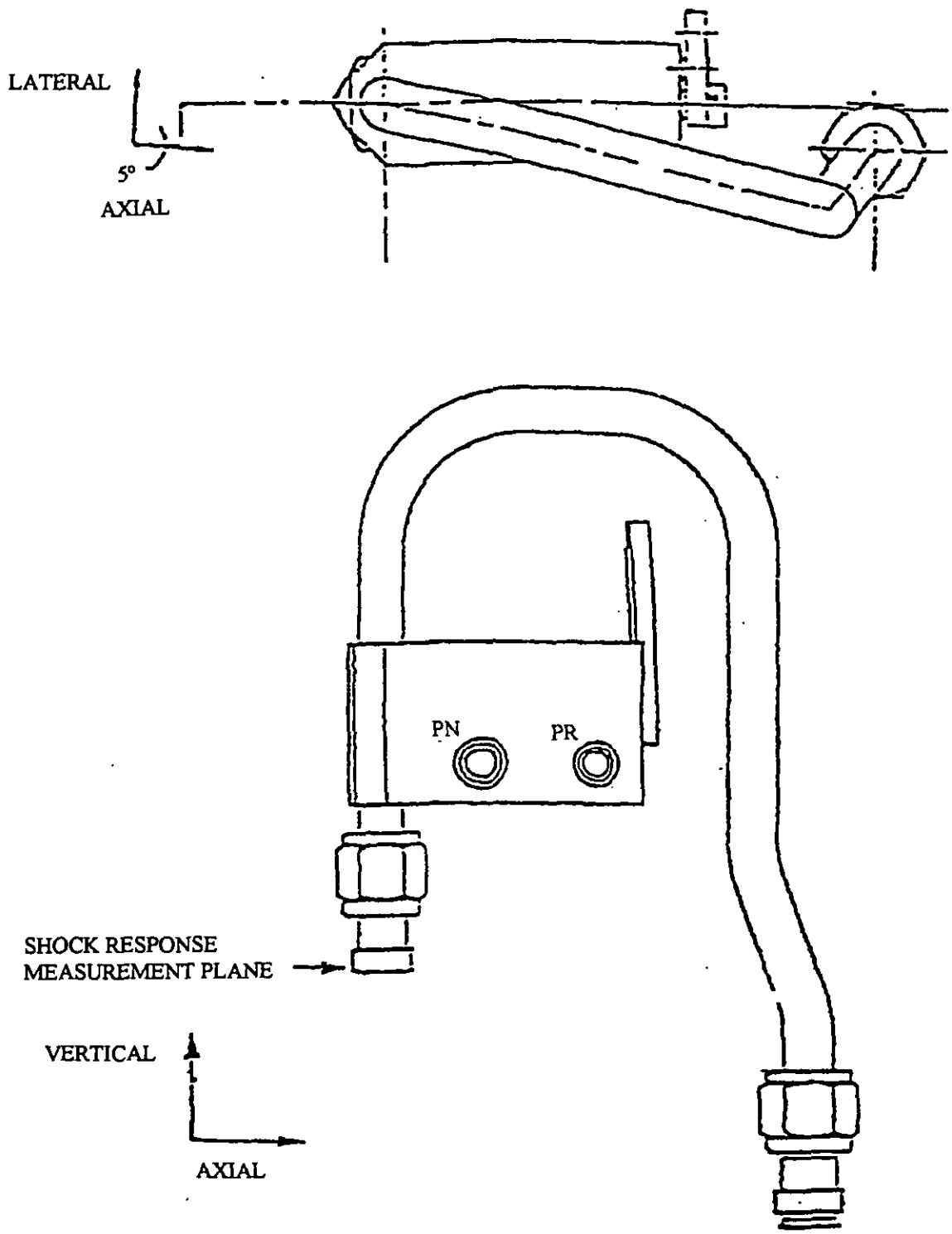


FIGURE 5. Shock test axes.

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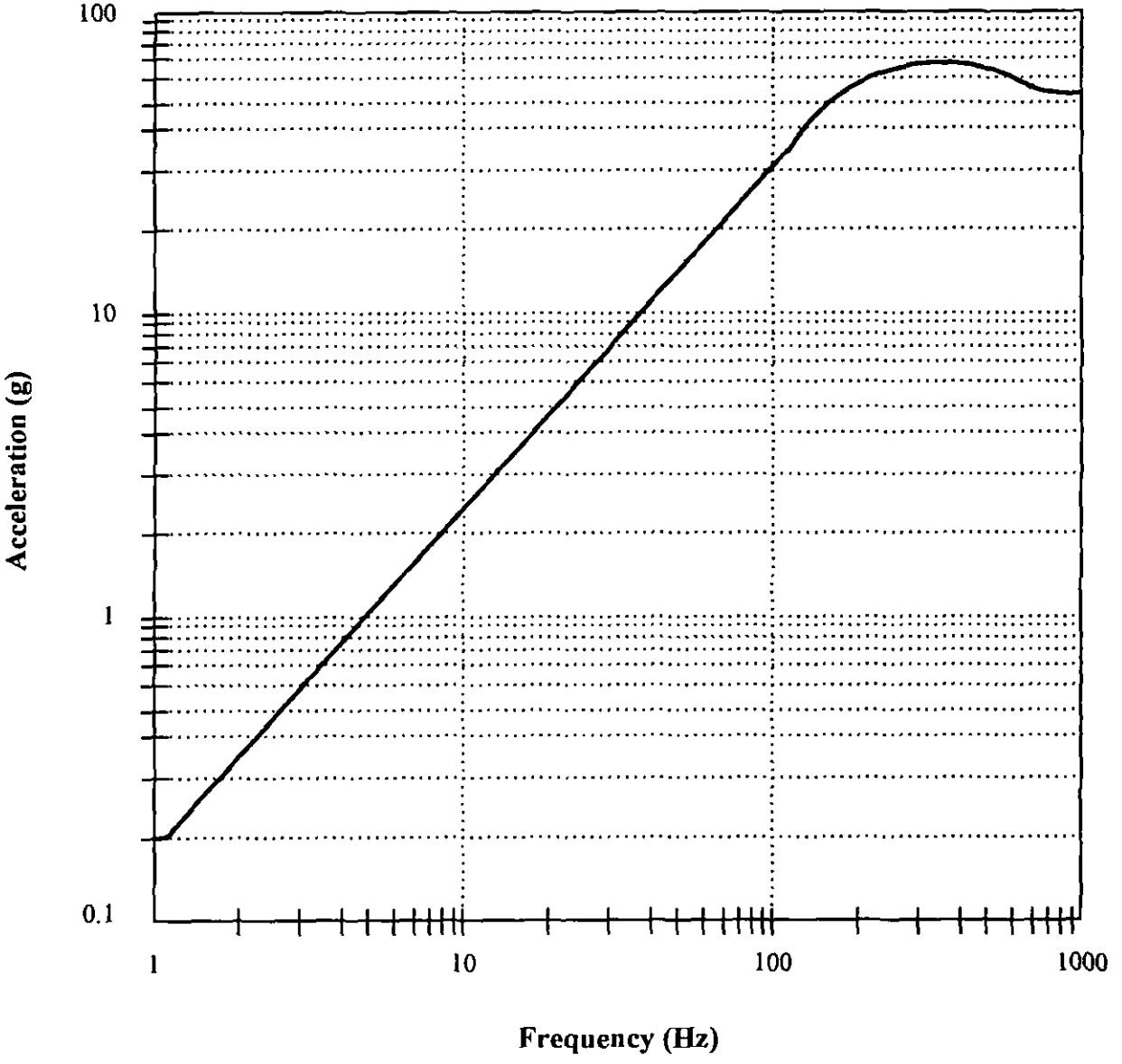


FIGURE 6. Gun firing shock-vertical.

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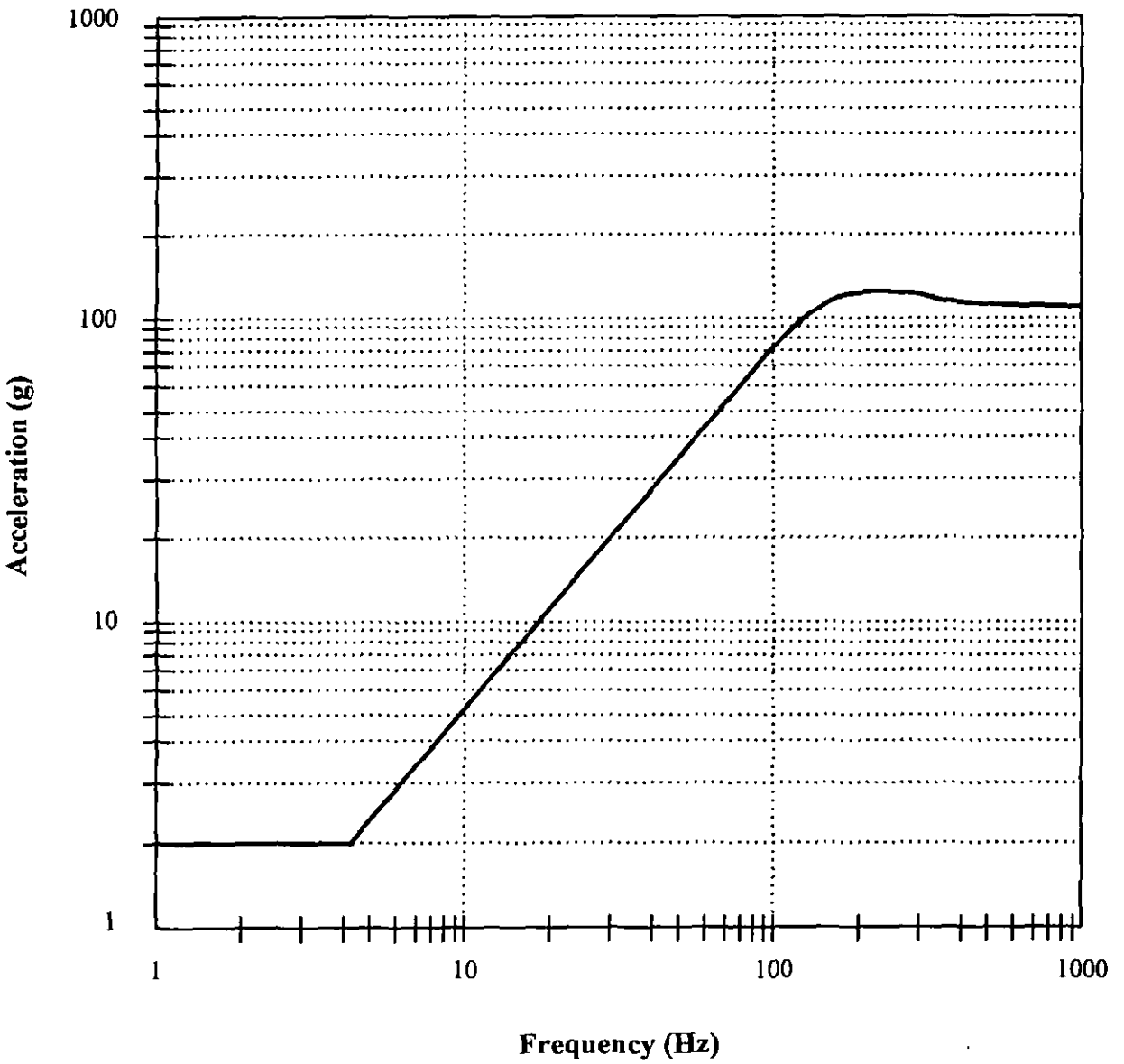


FIGURE 7. Gun firing shock-axial.

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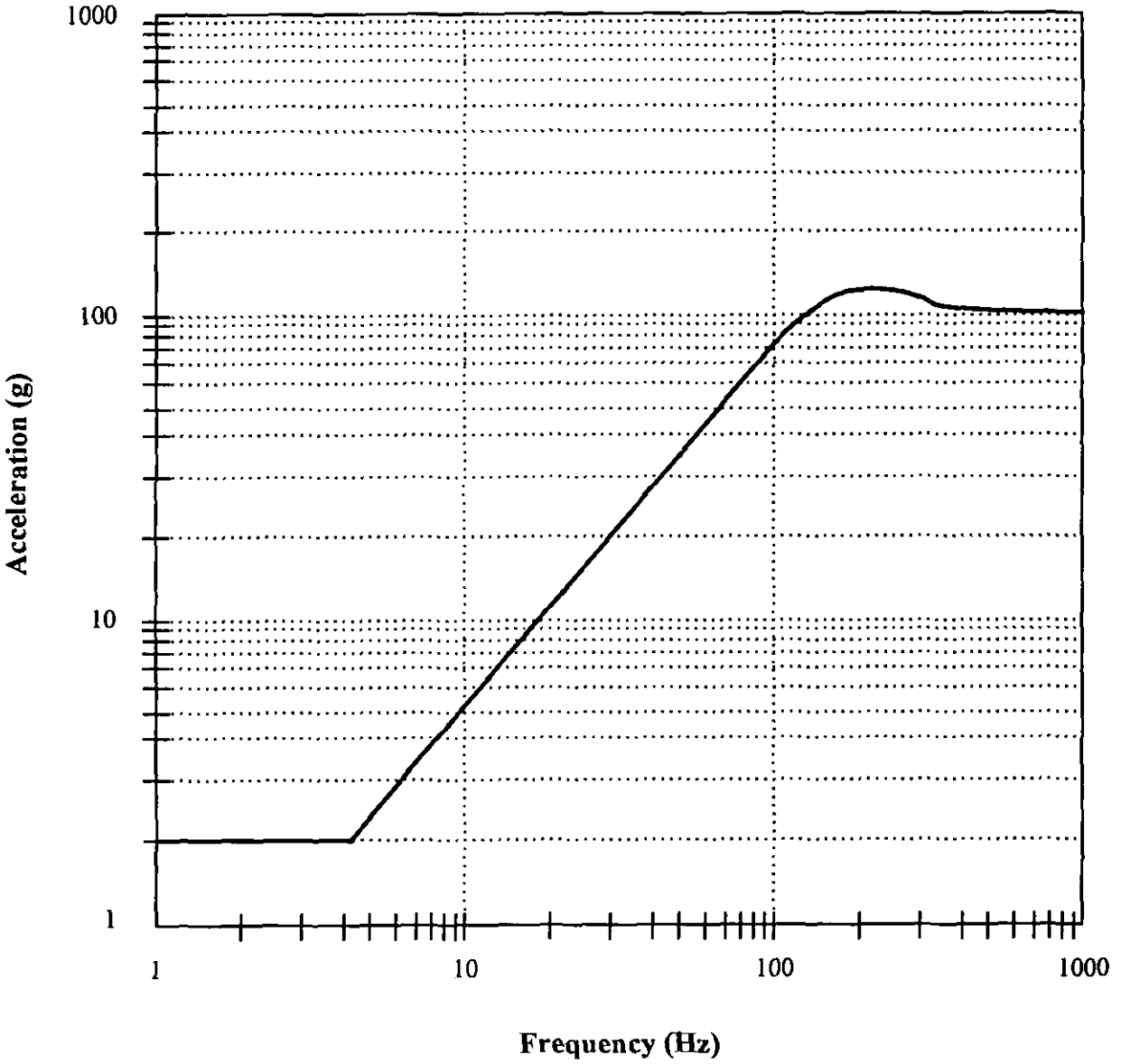


FIGURE 8. Gun firing shock-lateral.

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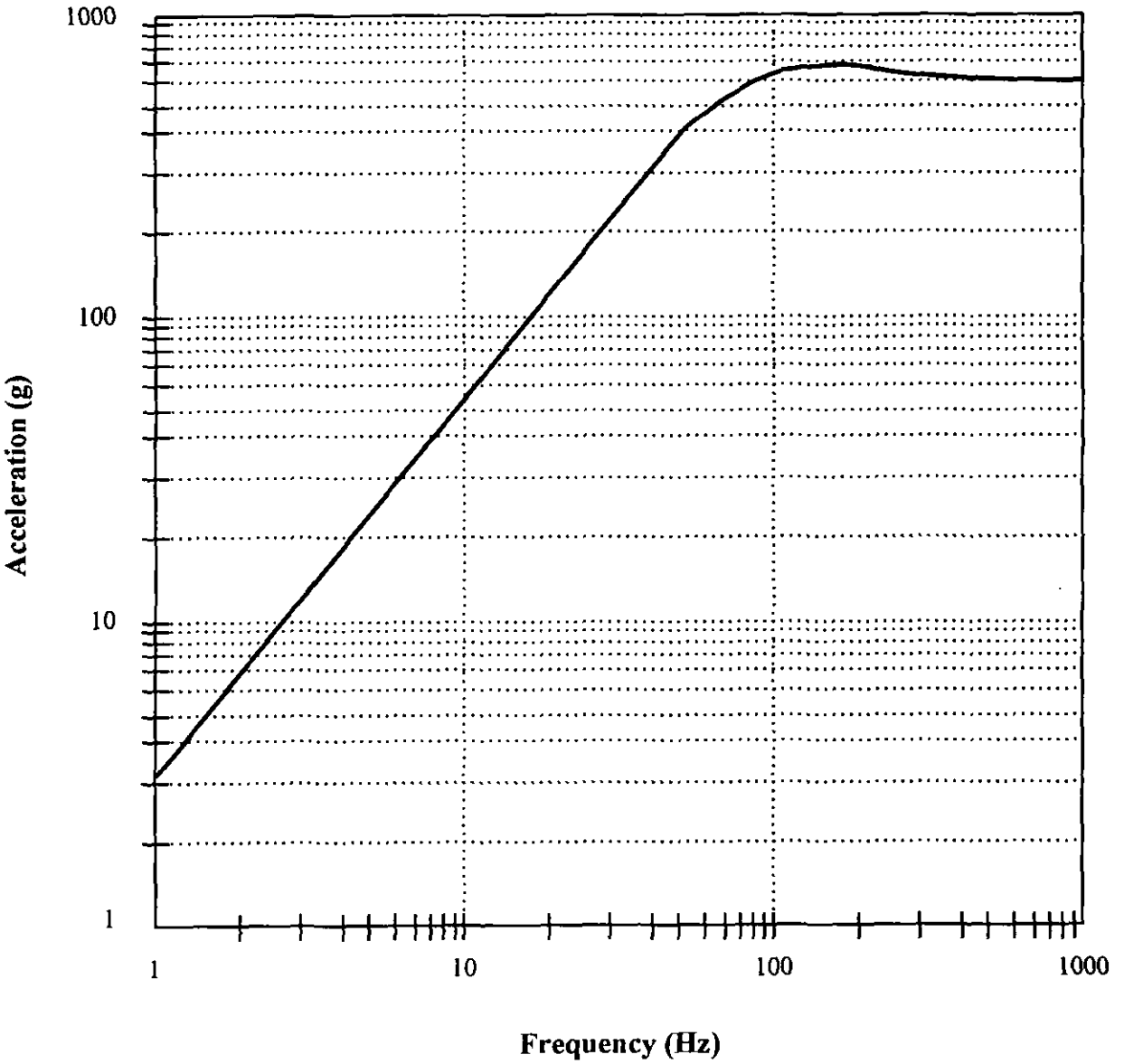


FIGURE 9. Ballistic shock-vertical.

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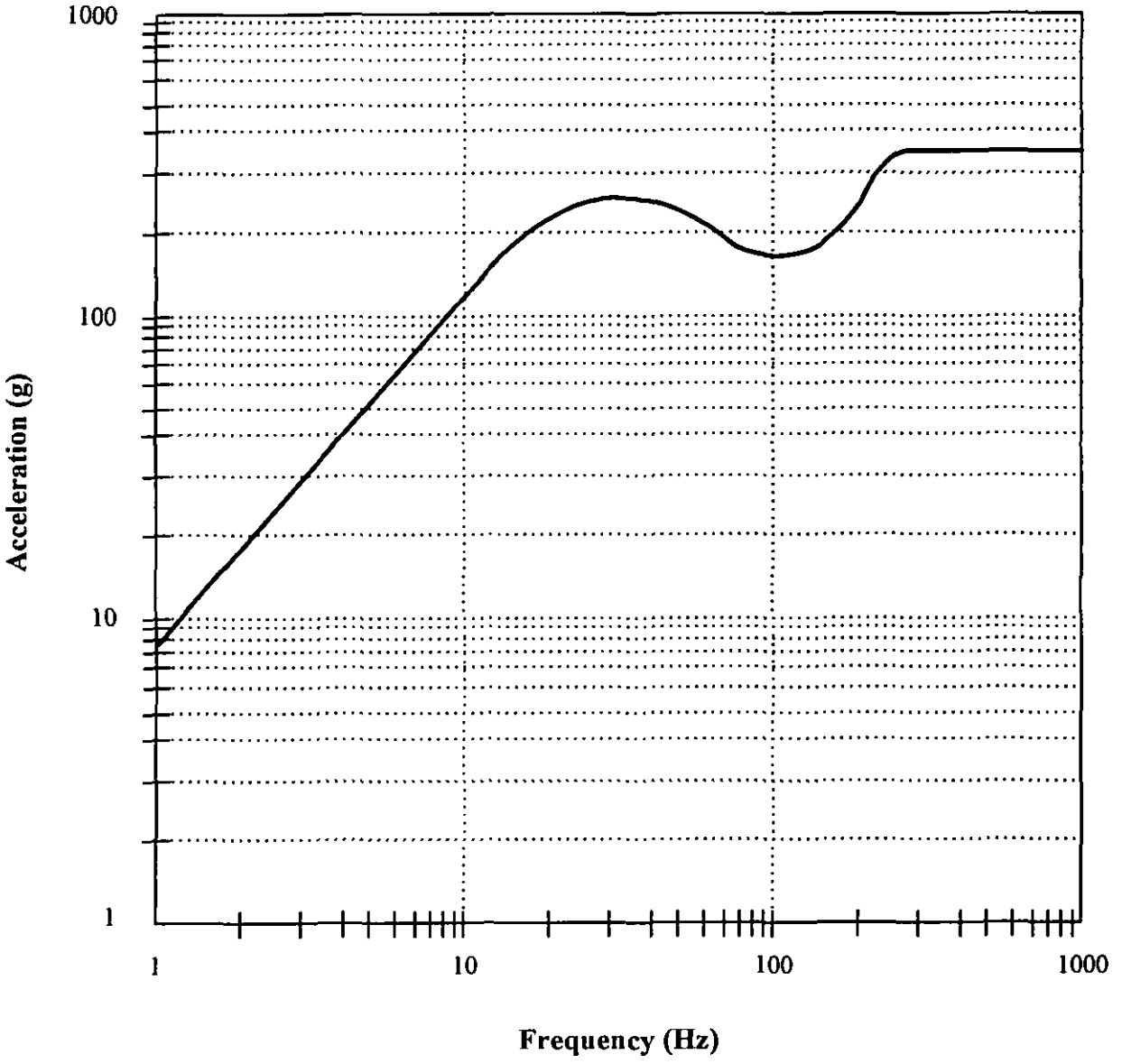


FIGURE 10. Ballistic shock-axial.

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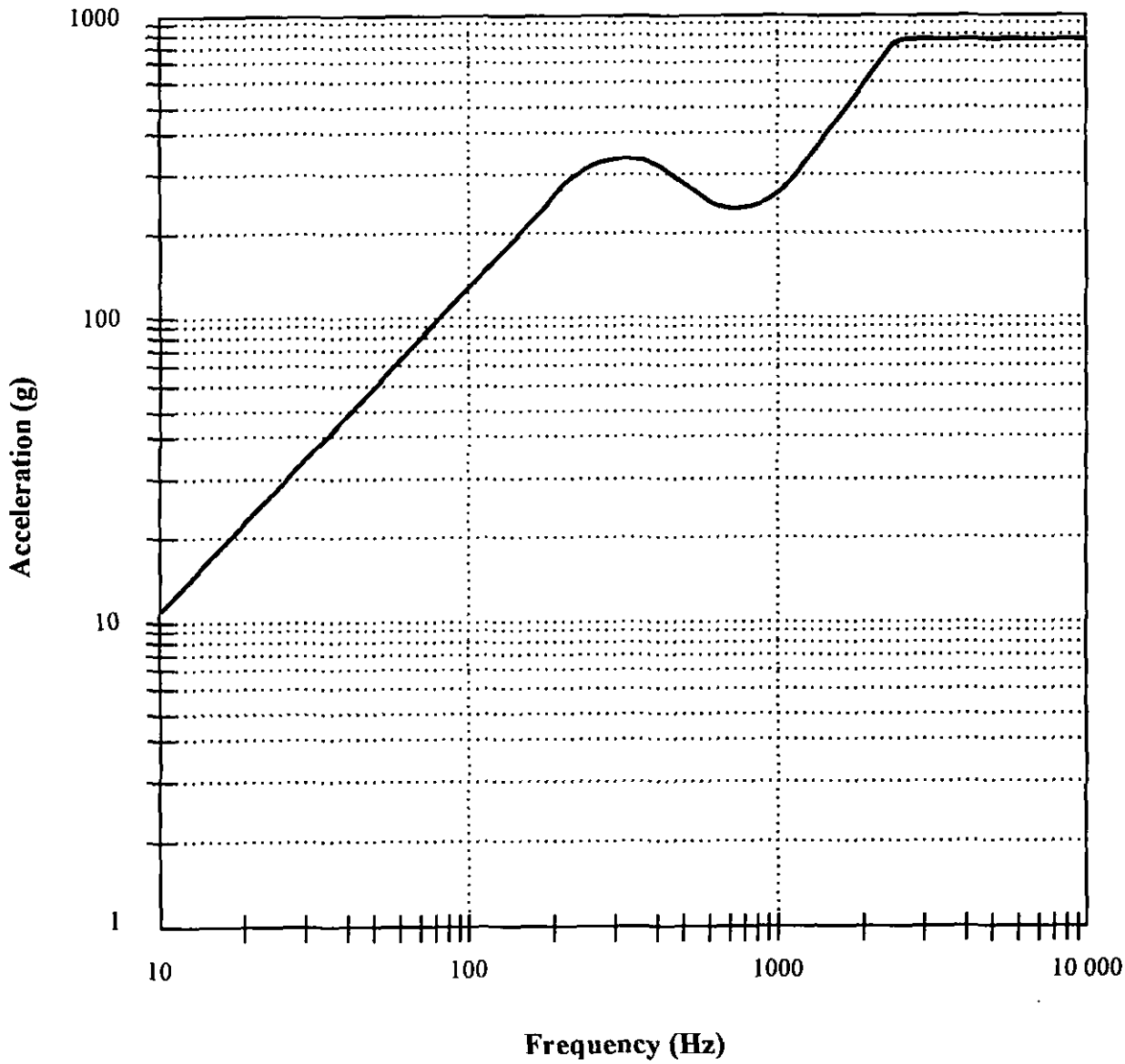
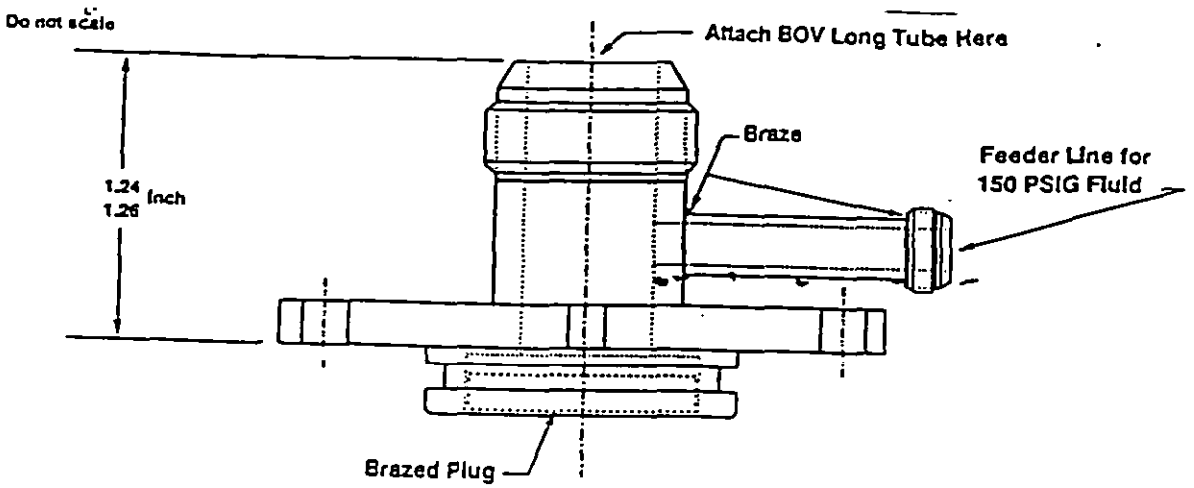


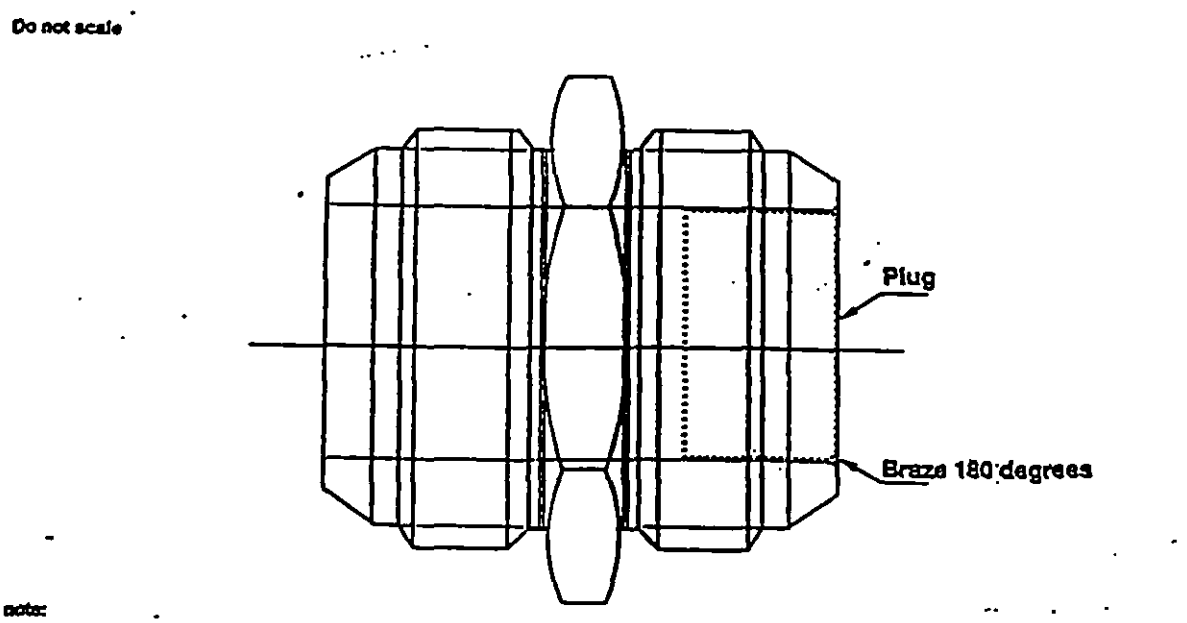
FIGURE 11. Ballistic shock-lateral.

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notes:
Make from Bendix Filter Inlet Fitting P/N 2539721 or P/N 2540333.
Install In The Long Tube Where BOV Attaches To The HMU Fuel Filter.

FIGURE 12A. Two bolt flange fitting with a brazed plug to prevent pressurization of the HMU and addition of a small feeder line.



note:
Install at BOV Inlet (Short Tube End at the Fuel Pump Outlet)

FIGURE 12B. An 815-8S fitting with a brazed plug to prevent pressurization of the HMU.

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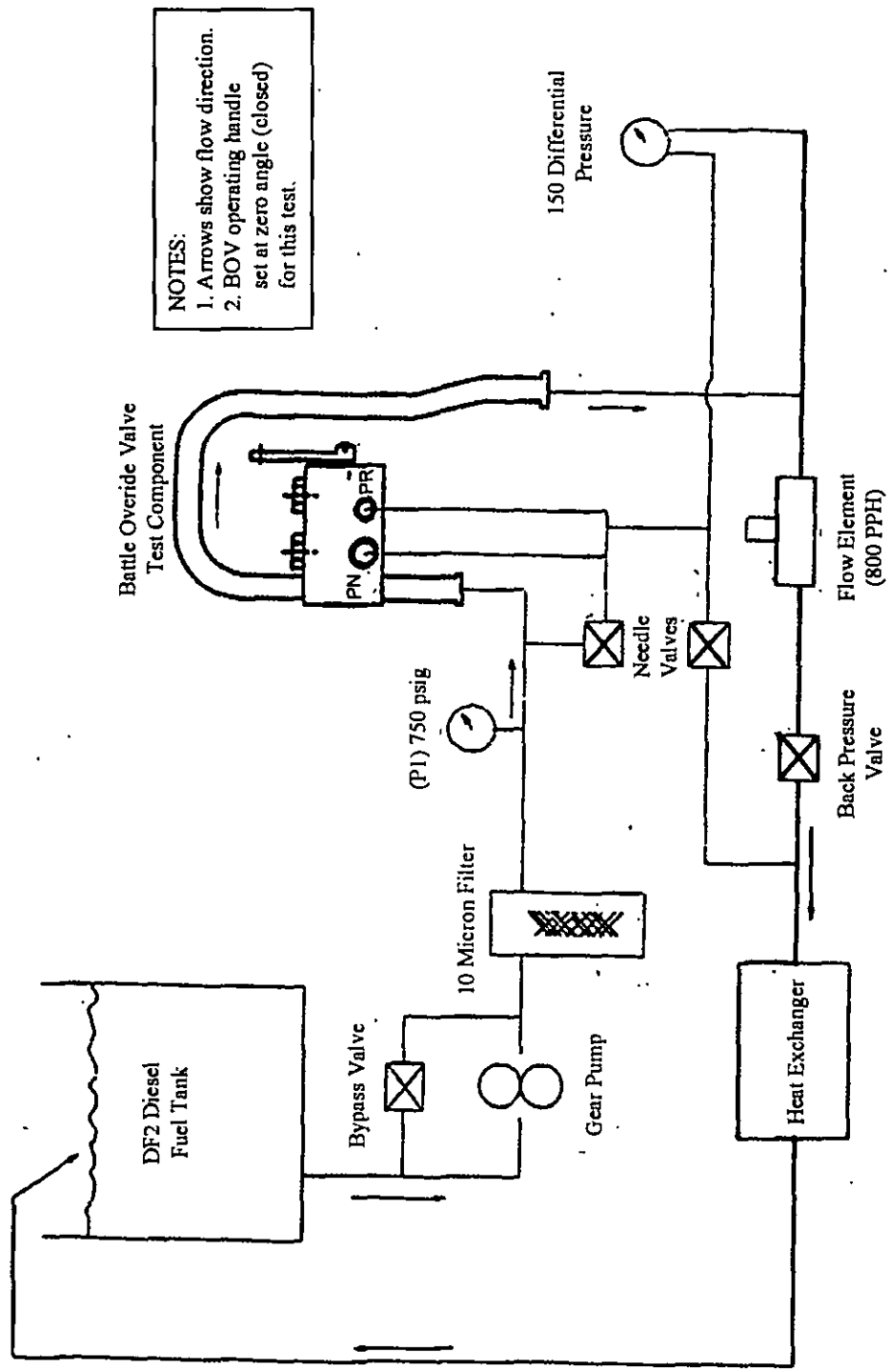


Figure 13. BOV schematic, operation cycle testing.

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VALVE BYPASS, FUEL, MECHANICAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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